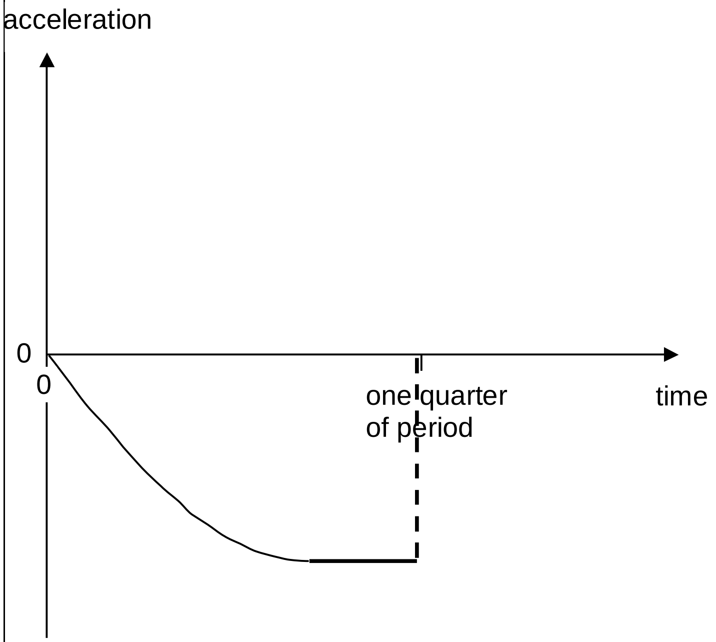
	$= 1.56 \times 10^3$ Bq	
9(a)	Simple harmonic motion is defined as an oscillatory motion in which the acceleration of an object is directly proportional to the displacement of the object from its equilibrium position, and the acceleration is always directed towards that position.	[1] [1]
(b)(i)	– 9.81 m s^{-2} When the plate is accelerating downward at 9.81 m s^{-2}, the sand will be undergoing acceleration of free fall. At this acceleration, the gravitational force acting on the sand particle is just sufficient to cause the acceleration of the sand particle and thus normal contact force equals zero.	[1] only accept negative answer [1] [1]
(b)(ii)1.	Gradient of graph = $-\omega^2$ $-\dfrac{12.5}{2.5\times10^{-2}} = -\omega^2$ $-\dfrac{12.5}{2.5\times10^{-2}} = -(2\pi f)^2$ $f = 3.56$ Hz	[1] correct gradient [1] correct link from gradient to *f* [1] correct answer
(b)(ii)2.	$\text{KE} = \dfrac{1}{2}mv^2$ $= \dfrac{1}{2}m\left(\omega\sqrt{x_0^{\,2} - x^2}\right)^2$ $= \dfrac{1}{2}(1.2\times10^{-5})(500)(0.025^2 - 0.010^2)$ $= 1.58\times10^{-6}$ J	[1] correct substitution [1] correct answer
(b)(ii)3.	$x = x_0 \sin(2\pi f t)$ $2.0 = 2.5\sin(2\pi(3.56)t)$ $t = 0.0415$ s	[1] for substitution [1] for answer Can accept if use x = 1.95

(b)(iii)		[1] correct shape [1] negative
(c)(i)	At the equilibrium position, the upward tension force is equal to the downward weight. When the pendulum is displaced leftwards along a circular arc, the component of the weight tangent to the arc acts towards the right and is parallel to its displacement. This component provides the restoring force At equilibrium position, the upthrust equals to weight. As the block is displaced downwards from its equilibrium position, the volume of water displaced increases and hence, upthrust increases. The restoring force for the floating block is the resultant force due to the upthrust on the block by the water and the block's weight.	[1] [1] [1] [1]
(c)(ii)	Resonance Driving frequency of the water waves is equal to the natural frequency of the oscillation of the block.	[1] [1]
10(a)(i)		[1] for correct